

DSA0502 – QUERY PROCESSING FOR DATA SCIENCE

LAB PROGRAMS (11-12)

NAME: B KIRTHANAA
192324244

REG. NO -

11. Create a DataFrame of ten rows and four columns with random values. Convert some values to NaN values. Write a Pandas program to highlight the NaN values.

Aim:

To create a DataFrame with random values, introduce missing (NaN) values, and highlight the missing values using Pandas.

Code:

```
import pandas as pd
import numpy as np

df = pd.DataFrame(np.random.randint(1, 100, (10, 4)),
                  columns=["A", "B", "C", "D"])

# Convert some values to NaN
df.iloc[2, 1] = np.nan
df.iloc[5, 3] = np.nan
df.iloc[7, 0] = np.nan
df.style.highlight_null(color="red")
```

OUTPUT:

```
[2]
✓ Os
#Q11
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame(np.random.randint(1, 100, (10, 4)),
                  columns=["A", "B", "C", "D"])

# Convert some values to NaN
df.iloc[2, 1] = np.nan
df.iloc[5, 3] = np.nan
df.iloc[7, 0] = np.nan

# Highlight NaN values
df.style.highlight_null(color="red")
```

	A	B	C	D
0	28.000000	67.000000	89	50.000000
1	20.000000	35.000000	70	43.000000
2	53.000000	nan	21	27.000000
3	98.000000	9.000000	43	43.000000
4	88.000000	50.000000	82	81.000000
5	40.000000	16.000000	99	nan
6	85.000000	48.000000	59	42.000000
7	nan	18.000000	99	19.000000
8	36.000000	57.000000	12	47.000000
9	47.000000	10.000000	86	34.000000

12. Create a DataFrame of ten rows and four columns with random values. Write a Pandas program to set the DataFrame background color to black and the font color to yellow.

Aim:

To create a DataFrame and apply custom background and font colors using Pandas styling.

Code:

```
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame(np.random.randint(1, 100, (10, 4)),
                  columns=["A", "B", "C", "D"])

# Apply style
df.style.set_properties(**{
    "background-color": "black",
    "color": "yellow"
})
```

OUTPUT:

[3]
✓ 0s

```
#Q12
import pandas as pd
import numpy as np

# Create DataFrame
df = pd.DataFrame(np.random.randint(1, 100, (10, 4)),
                  columns=["A", "B", "C", "D"])

# Apply style
df.style.set_properties(**{
    "background-color": "black",
    "color": "yellow"
})
```

▼

```
...
   A  B  C  D
0  21 93 92 75
1  87 40  1 41
2  51 35 25 10
3   1 97 57 96
4  82 37 83 81
5  63  5 80 47
6  93 73 72 27
7  71 39 33 52
8  64 18 78 52
9  44 91 79 38
```